

KUNAL KARANKAL

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Objective

Detail-oriented Mechanical Engineer with 2+ years of experience in R&D, 3D modeling, Drafting and CAD design. Adept at delivering high-quality designs using Inventor, Creo, and SolidWorks. Focused on innovation and continuous improvement, with a strong curiosity for how designs work and a drive to refine them through better methods and engineering insight.

Education

Shri Vile Parle Kelavani Mandal's Institute of Technology, Dhule

B.Tech in Mechanical Engineering, CGPA: 8.72

Sep. 2020 – May 2023

Dhule, Maharashtra

R. C. Patel Polytechnic, Shirpur

Diploma in Mechanical Engineering, 83.90%

Sep. 2017 – May 2020

Shirpur, Maharashtra

Experience

Pall Corporation

Design Engineer

Nov. 2025 – Present

Pune, Maharashtra

- Assisted in development of Configurator automation tools using VB.Net to generate ready-to-use models from input parameters, significantly streamlining engineering workflows and reducing design cycle time.
- Managed & arranged data to optimize productivity. Reducing working time with sorted data.
- Participated in VAVE strategies training and performed tasks to identify problems, solutions & possible optimization.
- Collaborated with core R&D team & sales team to boost sales & reduce working cycle

CG Power and Industrial Solutions Ltd

R&D Design Engineer

Dec. 2023 – Nov. 2025

Nashik, Maharashtra

- Designed switchgear assemblies including RMU Cladding and copper connections using Autodesk Inventor and AutoCAD.
- Executed retrofitting of AIS, designing adaptation kits to upgrade legacy systems and extend product lifecycle.
- Engineered sheet-metal cladding and detailed fabrication designs, generating flat patterns and bend sequences for production.
- Develop prototypes and coordinate fabrication; perform fit, form, and function validation and designs based on feedback.
- Conduct product benchmarking and competitive analysis to drive design improvements and cost / manufacturability optimizations.
- Prepare detailed manufacturing instructions (MI) and apply GD&T for tolerance control, assembly checks, and quality assurance.

CG Power and Industrial Solutions Ltd

Intern

Sep. 2021 – Oct. 2021

Nashik, Maharashtra

- Studied material properties and finishes for switchgear, proposing optimal processes for stainless steel hardening.
- Performed 3D modeling and component analysis in Creo to support design validation and improve manufacturability.

Projects

Design, Fabrication, and CFD Analysis of Fin Configurations

May 2023

- Improved heat transfer coefficient by 15% through optimized fin geometry using Ansys and experimental validation.
- Modeling with Creo&Solidworks, Drafting, Validation, & Experimentation with Ansys Fluent
- Published findings in a Scopus-indexed Springer book chapter.

IoT Based Energy Management System

Jun. 2022

- Developed an app reducing energy evaluation time using Arduino, Android Studio, and PyCharm, with real-time graphical reports.

Technical Skills

CAD/Modeling: Creo, SolidWorks, NX, Inventor, AutoCAD, CATIA, Casting Modeling, Interlock Design

Simulation: Ansys, HyperWorks, MATLAB (Basic)

Design: GD&T, Parametric Modeling, Tolerance Analysis, Sheet Metal Design, 2D Drafting, Benchmarking

Tools: Part Families, Expressions, Wave Linking, Assemblies, ISO Standards, VB.Net

Languages: English, Hindi, Marathi, Japanese (Basic)

Publications

Book Chapter: “Computational Analysis of Various Fin Configurations: A Comprehensive Assessment,” *Advancements in Materials Processing Technology, Volume 1* (Springer, Scopus-indexed), 2023.